Part Three - A game that can end

Last meeting your character started scoring points. Today it gets something to avoid, and your game gets a way to end.

If your sketch from last time is broken, the Canvas page has a Day 1 checkpoint. Paste it into your main tab, keep your own `MyCharacter` and `Cloud` tabs, and you're caught up.

Step 7 - A hazard

Build it. Add one more Blob, in the same three places as everything else. Declare `Blob hazard;` at the top. Create it in `setup()`:

```
// The hazard starts in the top-left corner, away from the player
hazard = new Blob(this, 80, 80, 22, 3, 2, color(220, 60, 60));
```

In `draw()`, move and draw it, right after the coin.

It starts in a corner on purpose. A hazard made right on top of your player would end the game the instant it started.

Step 8 - True or false

Build it. At the top of your sketch, add:

```
boolean gameOver = false;   // true or false, starts false
```

> **Aside: `boolean`**
>
> A **`boolean`** is the simplest variable there is. It's `true`, or it's `false`, and that's all. Here it remembers one thing: is the game over?

Then, after the coin's `if`, add one for the hazard:

```
if (player.collidesWith(hazard)) {
  gameOver = true;
}
```

Run it, and touch the hazard.

> **Check.** Nothing seems to happen. The game keeps going.

The variable changed. `gameOver` is `true` now, and nothing in your sketch reads it yet. The next two Steps give it a reader.

Step 9 - A function of your own

`draw()` is getting long. Before it gets longer, you're going to give the game part of it a name of its own.

Build it. Below the closing brace of `draw()`, make a new function called `playGame()`. Move into it your coin, your hazard, your player, both collision `if`s, and the score display. Leave the sky, clouds, and ground in `draw()`.

```
void playGame() {
  coin.move();
  coin.drawSprite();

  hazard.move();
  hazard.drawSprite();

  player.followMouse();
  player.drawSprite();

  // Touching the coin:  score a point, and the coin jumps somewhere new
  if (player.collidesWith(coin)) {
    score = score + 1;
    coin.moveTo(random(width), random(100, height - 150));
  }

  // Touching the hazard:  the game is over
  if (player.collidesWith(hazard)) {
    gameOver = true;
  }

  fill(50);
  textSize(24);
  text("Score: " + score, 20, 36);
}
```

In `draw()`, after the scenery, call it:

```
  playGame();
```

Aside: writing your own function

`void playGame() { ... }` makes a function, the same shape as `setup()` and `draw()`. The difference is who calls it. Processing calls `setup()` and `draw()` for you. This one runs when your sketch says its name: `playGame();`. You can name a function anything, so name it what it does.

Run it.

Check. Everything looks exactly the same as before.

That means it worked. Moving code into a function of its own is called **refactoring**, and a refactor that leaves the screen exactly the same is a success.

Step 10 - Two screens

Build it. Below `playGame()`, add a second function, for the end of the game:

```
void showGameOver() {
  textAlign(CENTER, CENTER);
  fill(220, 60, 60);
  textSize(52);
  text("Game Over", width/2, height/2 - 20);
  fill(50);
  textSize(22);
  text("Final score: " + score, width/2, height/2 + 26);
  textAlign(LEFT, BASELINE);   // put text alignment back to normal
}
```

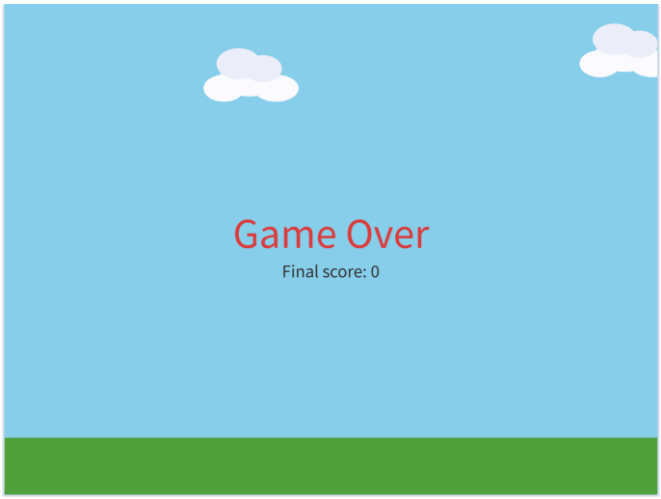
Then, in `draw()`, replace `playGame();` with this:

```
  if (!gameOver) {
    playGame();       // still playing
  } else {
    showGameOver();   // it's over
  }
```

> **Aside: `!` and `else`**
>
> `!` means **not**. So `!gameOver` reads *not game over*: still playing. `else` is the other half of an `if`. When the condition is true, the first braces run. When it's false, the braces after `else` run instead. Exactly one of the two runs, every frame.

Run it, and touch the hazard.

> **Check.** The game stops, and *Game Over* comes up with your final score.



Game over. The sky, clouds, and ground are still there, because they're still in `draw()`.

This is the big idea of today. Your game is always in a **mode**. Right now there are two: playing, and game over. The `boolean` remembers which mode you're in, and the `if` / `else` in `draw()` picks which one to run.

Build it. Test the other branch by cheating. Change `boolean gameOver = false;` to `true`, and run.

> **Check.** The game opens straight onto the game over screen, with a score of 0.

Set it back to `false`.

6.5 Comparing numbers

A condition can compare numbers, too:

```
if (score >= 10) {
  background(255, 0, 0);   // only happens once score is 10 or more
}
```

You write	It asks
`score > 10`	greater than 10?
`score < 10`	less than 10?
`score >= 10`	10 or more?
`score <= 10`	10 or less?
`score == 10`	exactly 10?

Watch out. `==` is a question: *are these equal?* A single `=` is a command: *make this equal to that*. Write `if (score = 10)` by mistake, and the sketch stops with:

```
Type mismatch: cannot convert from int to boolean
```

An `if` needs a yes or a no, and `score = 10` is a command. Make it `==`.

Part Four - Make it yours

Your turn. Rewrite the game over screen so it sounds like your game: different words, different colors.

> **Watch out.** Keep the last line, `textAlign(LEFT, BASELINE);`. It puts text back to normal after the centered game over text, and without it every `text()` after it comes out centered too.

Your turn. Tune the hazard: its speed, its size, where it starts. A harder game, or an easier one?

Your turn. Optional: any of these.

- A second hazard, `hazard2`, with its own `if`. Give it a different speed or size.
- A second coin, `coin2`, worth more: `score = score + 5;`
- Change what ends the game. A hazard? Reaching a score of 20? Both?
- A background that changes as the score goes up. Stack a few `if`s right after the sky, one for each score, so the later ones paint over the earlier:

```
if (score >= 5) {
  background(255, 170, 80);    // sunset
}
if (score >= 10) {
  background(30, 30, 70);      // night
}
```

Your turn. Optional: give your player lives. Start with `int lives = 3;` at the top. When the hazard hits, take one away, and move the hazard somewhere else, exactly like the coin. Set `gameOver` to `true` only when `lives` reaches `0`. Show the lives on the screen, under the score.

Then play your game. This is where it becomes one, so leave time to play it.

Chapter summary

You started with a coin that did nothing when you touched it. You end with a game: a score that counts, something to avoid, and a screen that says it's over.

You built	It's an example of
`player.collidesWith(coin)`	a yes-or-no question, using the **hitbox** from Chapter 2
`if (...) { ... }`	doing something only when a **condition** is true
`int score = 0;` at the top	a variable every function can see
`text("Score: " + score, 20, 36)`	words and a number, glued together, on the screen
`score = 7`, then back	testing by cheating
the racing score, and `moveTo()`	`draw()` asks every frame, so the fix is to end the touch
`boolean gameOver`	a `true` or `false` that remembers which mode you're in
`playGame()`	**refactoring** into a function of your own
`if (!gameOver) { ... } else { ... }`	two modes, and `draw()` picking one

New syntax in this chapter

Syntax	Where you met it
`collidesWith()`	6.1
`if (condition) { }`	6.2
`int`	Step 3
`fill()`, `textSize()`, `text()` for words	Step 4
`+` between text and a number	Step 4
`score = score + 1`	Step 5
`boolean`, `true`, `false`	Step 8
`void playGame() { }`, and calling it	Step 9
`!`, `else`	Step 10
`textAlign()`	Step 10
`>`, `<`, `>=`, `<=`, `==`	6.5

Vocabulary

`boolean` -- a variable that's only ever `true` or `false`. (Step 8)

condition -- the yes-or-no question in the parentheses of an `if`. (6.2)

hitbox -- the invisible circle that decides whether something is touching. For your character, it's the radius in the `super(...)` line. (6.1)

`if` statement -- "if this is true, do this block of code." (6.2)

`int` -- a variable that holds a whole number. (Step 3)

mode -- which part of the game your program is in right now: playing, or game over. (Step 10)

refactoring -- moving code around so it's easier to read, while it does exactly the same thing. (Step 9)

Check your understanding

1. What does `collidesWith()` hand back? What decides whether two things are touching?
2. Why does `int score = 0;` go at the very top of the sketch?
3. Before you added `moveTo()`, why did the score race upward?
4. What's the difference between an `int` and a `boolean`? Give an example of each from your own game.
5. Why did touching the hazard in Step 8 seem to do nothing?
6. `=` and `==`. Which one asks a question?

What's next

Your game can end. Once it's over, though, the only way to play again is to press Stop and Run.

In Chapter 7 you finish it. A way to win as well as lose, a click to play again, and three meetings to make it a game somebody else wants to play.